

1. Let $\mathbf{v} = \langle a, b, c \rangle$ and let α , β , and γ be the angles between $\mathbf{v}$ and the positive x -axis, the positive y -axis, and the positive z -axis, respectively (see the figure below).
- Prove that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$.
 - Find a vector that makes a 45° angle with $\mathbf{i}$ and $\mathbf{j}$. What angle does it make with $\mathbf{k}$?
 - Is there a vector that makes a 30° angle with $\mathbf{i}$ and $\mathbf{j}$? Explain.
 - Find a vector $\mathbf{v}$ such that $\alpha = \beta = \gamma$. What is the angle?

